

Checklist:

Engineering Management Systems & Tools

1. Use systems thinking for management problems

- ☐ Identify the **stocks** in the system: accumulated quantities such as managers, trained managers, pull requests, incidents, technical debt, or available time.
- ☐ Identify the **flows** that change those stocks: hiring, training, departures, coding, review, deployment, recovery, debugging.
- ☐ Label every flow as a **rate**, not a quantity.
- ☐ Look for **feedback loops** where downstream outcomes affect upstream work.
- ☐ Separate **causal relationships** from mere correlations.
- ☐ Identify which stock or flow is currently the real constraint.
- ☐ Avoid optimizing a rate that does not matter because its downstream stock is already empty or blocked.
- ☐ Use the model to test hypotheses before committing to work.

2. Manage product work through exploration, selection, and validation

Problem discovery

- ☐ Explore users' pain points: what problems do they actually experience?
- ☐ Explore users' purpose: what are users trying to accomplish?
- ☐ Benchmark against similar companies or industries.
- ☐ Look for hidden cohorts with unusual or surprising needs.
- ☐ Identify areas where your company has unusual strengths.
- ☐ Identify competitive moats or advantages.
- ☐ Look for compounding leverage: work that becomes more valuable over time.

Problem selection

- ☐ Ask what is needed to **survive the current round**.
- ☐ Ask what is needed to **survive the next round**.
- ☐ Ask what would help you eventually **win rounds**, not only survive them.
- ☐ Consider different time frames: six months, two years, five years.
- ☐ Consider industry trends and where they are heading.
- ☐ Estimate return on investment, especially for quick or easy wins.
- ☐ Identify experiments that will make future selection easier.

Solution validation

- ☐ Write a customer letter or launch announcement before building.
- ☐ Identify prior art and how peers have approached the problem.
- ☐ Find reference users who are willing to be early users.
- ☐ Prefer experiments over analysis when validation is cheap.
- ☐ Find the fastest path to learning whether the solution works.
- ☐ Test switching costs for users moving to the solution.
- ☐ Avoid building the whole solution merely to validate it.

3. Create strategies and visions

Strategy

- ☐ Define the specific challenge the strategy addresses.
- ☐ Write a diagnosis that clearly explains the challenge.
- ☐ Identify constraints and trade-offs.
- ☐ Define guiding policies for addressing the challenge.
- ☐ Translate policies into concrete actions.
- ☐ Keep the strategy practical, specific, and detailed.
- ☐ Write as many strategies as are useful, but avoid unnecessary documents.

Vision

- ☐ Define the future state where today's trade-offs are no longer mutually exclusive.
- ☐ Write a short vision statement.
- ☐ Define the value proposition for users and the company.
- ☐ Identify capabilities needed to deliver the value proposition.

- ☐ Name current constraints that the future state resolves.
- ☐ Name future constraints that may appear.
- ☐ Link relevant plans, metrics, and reference materials.
- ☐ Synthesize the vision into a short narrative.
- ☐ Test the document with others and iterate.
- ☐ Refresh the vision periodically.
- ☐ Use present tense.
- ☐ Write simply.
- ☐ Prefer one unified vision for each complete area.

4. Define useful metrics and goals

- ☐ Ensure every goal has a **target state**.
- ☐ Include a **baseline** showing where you are today.
- ☐ Include a **trend** showing current velocity.
- ☐ Include a **time frame** for the change.
- ☐ Check whether someone unfamiliar with the area can judge the goal's difficulty.
- ☐ Check whether success can be evaluated afterward.
- ☐ Distinguish **investment goals** from **baseline goals**.
- ☐ Pair investment goals with counterbalancing baseline metrics.
- ☐ Keep investment goals few enough to drive planning discussions.
- ☐ Carry baseline goals across planning cycles where useful.
- ☐ Treat some baselines as contracts, especially when tied to SLOs.

5. Use metrics to guide broad organizational change

- ☐ Start by exploring the data in a usable format.
- ☐ Dive into the largest contributors or drivers.
- ☐ Attribute each metric to the teams that can affect it.
- ☐ Build context around each team's performance.
- ☐ Benchmark teams against relevant peers.
- ☐ Nudge teams with contextual information, not just dashboards.
- ☐ Agree on baseline metrics with key teams.
- ☐ Review performance monthly or quarterly.
- ☐ Advocate for prioritization when teams lack agreed-upon baselines.
- ☐ Minimize top-down orchestration where self-correction is possible.

6. Run technical migrations well

Decide whether migration is needed

- ☐ Confirm the migration is the best path to reduce technical debt.
- ☐ Confirm the migration creates technical leverage or capacity.
- ☐ Recognize that large migrations often trade short-term contribution for long-term capacity.
- ☐ Avoid pretending technical debt can be solved only through isolated teamwork.

De-risk

- ☐ Write a design document.
- ☐ Test the design with teams likely to have the hardest migration.
- ☐ Iterate with edge cases and atypical patterns.
- ☐ Test against the next six to twelve months of roadmap.
- ☐ Embed with one or two challenging teams before broader rollout.
- ☐ Get endorsement from a team that completed the migration.

Enable

- ☐ Build self-service tooling where possible.
- ☐ Make the migration incremental and reversible.
- ☐ Document the process clearly.
- ☐ Test documentation with real teams.
- ☐ Prefer programmatic migration for the easy 90%.
- ☐ Improve tools and docs before scaling the migration.

Finish

- ☐ Stop the bleeding by ensuring new work uses the new approach.
- ☐ Avoid generating tracking tickets too early.
- ☐ Add visibility into migration status for teams and management.
- ☐ Push status to teams that need to migrate.
- ☐ Reserve recognition for successful completion, not only starting.
- ☐ Drive to 100% adoption when deprecating the legacy system.

7. Plan an engineering reorg

Validate that a reorg is the right tool

- ☐ Confirm the problem is structural.
- ☐ Make sure you are not avoiding a people-management issue.
- ☐ Confirm the problem already exists.
- ☐ Check whether the conditions are temporary.
- ☐ Avoid reorgs that merely work around broken relationships.

Project headcount

- ☐ Estimate an optimistic number based on what is barely possible.
- ☐ Estimate a natural-size number for each team and role if they were fully staffed.
- ☐ Estimate a realistic number based on historical hiring rates.
- ☐ Merge these into one planning number.
- ☐ Avoid overfitting the design to today's exact individuals.

Set manager-to-engineer ratio

- ☐ Decide whether managers are expected to do hands-on technical work.
- ☐ Use a lower ratio if managers do substantial technical work.
- ☐ Otherwise target roughly five to eight engineers per manager.
- ☐ Pick a ratio deliberately, probably in the six-to-eight range.

Define teams and groups

- ☐ Write a crisp mission statement for each team.
- ☐ Check whether each team would be exciting to join and manage.
- ☐ Put teams that need to work together close to each other.
- ☐ Define clear interfaces for each team.
- ☐ List ownership areas for each team.
- ☐ Create a gapless ownership map.
- ☐ Avoid implicit holes of ownership.
- ☐ Prepare compelling candidate pitches for each team.
- ☐ Check whether the design over-optimizes for individuals.

Staff the structure

- ☐ Identify people ready to fill roles now.
- ☐ Identify people who can grow into roles in time.

- ☐ Identify possible internal transfers.
- ☐ Identify external hires needed.
- ☐ Prefer people who already know the culture.
- ☐ Maintain a spreadsheet of every person, current team, and future team.
- ☐ Make sure nobody is accidentally omitted.

Commit and roll out

- ☐ Confirm the changes are meaningfully net positive.
- ☐ Check whether the structure will last at least six months.
- ☐ Name the problems discovered during design.
- ☐ Identify what might trigger the next reorg.
- ☐ Identify who will be most impacted.
- ☐ Explain the reasoning behind the reorganization.
- ☐ Document how each person and team will be impacted.
- ☐ Provide availability and empathy for impacted individuals.
- ☐ Speak privately with those most heavily impacted first.
- ☐ Prepare managers and key individuals to explain the reasoning.
- ☐ Send written documentation.
- ☐ Be available for discussion.
- ☐ Prefer small-group discussion over large all-hands when emotions are high.
- ☐ Double down on skip-level one-on-ones.

8. Identify your leadership controls

- ☐ Decide where you need to engage and where you should hang back.
- ☐ Choose controls appropriate to the team size and the leader-relationship.
- ☐ Use metrics to align on outcomes while preserving flexibility.
- ☐ Use visions to align on long-term direction.
- ☐ Use strategies to align on constraints and approaches.
- ☐ Use organization design to align wider sub-organizations.
- ☐ Use headcount and transfers as a prioritization mechanism.
- ☐ Use roadmaps to align on problem selection and solution validation.
- ☐ Use performance reviews to coordinate culture and recognition.
- ☐ Agree on the required degree of alignment for each control:
 - ☐ "I'll do it."
 - ☐ Preview.
 - ☐ Review.

- ☐ Notes.
- ☐ No surprises.
- ☐ Let me know.
- ☐ Establish clear interfaces with the leaders you support.
- ☐ Use inability to delegate as a signal that you may be micromanaging.

9. Build career narratives

- ☐ Avoid reducing career development to promotion ladders.
- ☐ Help people identify broad long-term goals.
- ☐ Identify gaps between current capabilities and future aspirations.
- ☐ Translate goals into projects, research, relationships, and skills.
- ☐ Use the manager's business context to find opportunities.
- ☐ Ask the individual to participate actively rather than outsource the thinking.
- ☐ Refresh the narrative quarterly or periodically.
- ☐ Prioritize goals for the next three to six months.
- ☐ Focus on disproportionate opportunities, not only contested career-path steps.

10. Communicate with media or public audiences

- ☐ Answer the question you want to be asked.
- ☐ Reframe difficult questions into ones you can answer constructively.
- ☐ Stay positive.
- ☐ Avoid being pulled into negative framing.
- ☐ Speak in threes.
- ☐ Repeat the same three concise points as your refrain.

11. Use Model, Document, Share

- ☐ Model the practice locally with your own team first.
- ☐ Measure team health and throughput before changing the process.
- ☐ Iterate until you are confident the approach works.
- ☐ Document the problem, learning process, and adoption details.
- ☐ Make the document specific enough for another team to adopt.

- ☐ Ask others to review the documentation for clarity.
- ☐ Share the approach and your rollout experience.
- ☐ Do not lobby for change prematurely.
- ☐ Avoid using this approach to circumvent organizational authority.
- ☐ Use it when slow adoption of a great approach is better than fast adoption of a good-enough one.

12. Design centralized decision-making groups

Decide whether to centralize

- ☐ Identify the consistency problem you are trying to solve.
- ☐ Decide whether the group should be authoritative or advisory.
- ☐ Understand which positive freedoms the group creates.
- ☐ Understand which negative freedoms the group removes.
- ☐ Avoid centralizing merely because inconsistency feels uncomfortable.

Design the group

- ☐ Define the group's influence.
- ☐ Define how teams interact with the group.
- ☐ Choose a size small enough to operate effectively.
- ☐ Decide expected time commitment.
- ☐ Decide whether members should identify primarily with the group.
- ☐ Define a structured selection process.
- ☐ Define term length.
- ☐ Decide whether terms are fixed or permanent.
- ☐ Decide how representative the group must be.
- ☐ Include the right functions, teams, tenures, and perspectives.

Watch failure modes

- ☐ Avoid domineering groups that reduce both positive and negative freedoms.
- ☐ Avoid bottlenecked groups that try to do too much.
- ☐ Avoid status-oriented groups whose value becomes recognition rather than contribution.
- ☐ Avoid inert groups that do little.
- ☐ Embed a manager who is responsible for iterating on the group's format.

13. Present to senior leadership

- ☐ Start with the conclusion.
- ☐ Tie the topic to business value.
- ☐ Explain why anyone should care in one or two sentences.
- ☐ Establish historical narrative: how things are going, how you got here, and what is planned next.
- ☐ Make an explicit ask.
- ☐ Provide a data-driven diagnosis.
- ☐ Show enough raw data for others to follow your analysis.
- ☐ State decision-making principles.
- ☐ Explain how principles lead to the proposed actions.
- ☐ Explain what is next and when it will be done.
- ☐ Return to the explicit ask at the end.
- ☐ Prepare for detours.
- ☐ Answer directly.
- ☐ Go deep enough in the data to handle unexpected questions.
- ☐ Discuss details when needed, but avoid getting trapped in irrelevant detail.
- ☐ Prepare a lot; practice a little.
- ☐ Make the goal of the meeting clear.

14. Manage time deliberately

- ☐ Run a quarterly time retrospective.
- ☐ Categorize the past three months of calendar time.
- ☐ Use the retrospective to adjust next quarter's allocation.
- ☐ Prioritize long-term success over short-term quality when trade-offs are necessary.
- ☐ Finish small, leveraged things.
- ☐ Stop doing work that should be dropped, recategorized, or delegated.
- ☐ Size work backward from available time.
- ☐ Decide how many hours can be spent, then fit the work to that limit.
- ☐ Delegate work "in the system," not only individual tasks.
- ☐ Build systems that allow others to handle the work.
- ☐ Trust the systems you build.

- ☐ Stop handling exceptions forever.
- ☐ Decouple meeting attendance from productivity.
- ☐ Do not assume attendance equals value.
- ☐ Hire before you are overwhelmed.
- ☐ Calendar-block focused work.
- ☐ Add several two-hour focus blocks each week.
- ☐ Get administrative support when complexity justifies it.

15. Build communities of learning

- ☐ Prefer facilitation over lecturing.
- ☐ Keep presentations brief.
- ☐ Leave most time for discussion.
- ☐ Use small breakout groups.
- ☐ Give each group a clear discussion prompt.
- ☐ Regroup and share learning from each breakout.
- ☐ Choose topics people already know about or encounter in daily work.
- ☐ Encourage senior or tenured people to attend.
- ☐ Provide optional pre-reads for people who dislike public first exposure.
- ☐ Start with a short check-in when useful.
- ☐ Ask each person for their name, team, and one sentence about what is on their mind.
- ☐ Keep the format lightweight and repeatable.
- ☐ Design the group so participants learn from each other, not only from the facilitator.